

Problem Statement Worksheet (Profitability Driver Analysis)

Within 6 weeks, analyze 10,000+ business transaction records to identify how factors such as quantity, unit price, discount rate, and cost influence revenue and profit.

The analysis will measure relationships using statistical techniques, identify the top 3–5 key drivers of profitability, and achieve a regression model with $R^2 \geq 0.70$, providing actionable insights to improve pricing and discount strategies.

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1 Context

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2 Criteria for success

- Data is cleaned and validated for analysis
- Key relationships between variables are identified
- At least 3–5 key drivers of profit are determined
- Statistical analysis explains a meaningful portion of profit variation ($R^2 \geq 0.70$)
- Clear and actionable business recommendations are provided

3 Scope of solution space

- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Correlation and regression analysis
- Revenue and profit driver analysis
- Discount impact analysis
- Data visualization
- Insight generation and recommendations

4 Constraints within solution space

- Analysis is limited to available dataset variables
- Data quality issues may affect accuracy
- Correlation does not imply causation
- Tool limitations for advanced analytics
- Results may not generalize beyond this dataset

5 Stakeholders to provide key insight

- Business Managers
- Sales Team
- Finance Team
- Pricing Strategy Team
- Senior Management

6 Key data sources

- Business Analytics Dataset (10,000+ transaction records)
- Key fields include:
 - Order details (Order_ID, Customer_ID)
 - Sales metrics (Quantity, Unit Price, Revenue)
 - Financial metrics (Cost, Profit)
 - Discount Rate
 - Payment Method

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Issue Tree- Capstone 2_Profitability Driver Analysis

